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Understanding Digital Image Fundamentals
From Raw Pixels to Optimized Storage Solutions

Today's Learning Objectives

Image Types: Binary, grayscale, color, indexed
Bit Depth: 1-bit, 8-bit, 16-bit, 24-bit systems
Color Models: RGB, CMYK, HSV representations
Resolution: Spatial vs. intensity resolution

Formats: BMP, JPEG, PNG, GIF, TIFF
Compression: Lossless vs. lossy techniques
Algorithms: Huffman, LZW, DCT, wavelets
Trade-offs: Quality vs. file size analysis

Real-world Applications:

- Web graphics, mobile photography
- Medical imaging systems
- Satellite imagery processing
- Social media platforms

From raw pixels to optimized storage solutions!

Digital Image Processing Pipeline

Raw Image Capture	Image Types & Storage	Enhancement	Analysis	Applications
Sensors CCD/CMOS	Formats Compression	Filtering Correction	Feature Detection	Recognition AI

Key Questions We'll Answer Today:

- How are different image types represented in memory?
- What are the advantages/disadvantages of each storage format?
- How do compression algorithms reduce file sizes?
- When should we use lossless vs. lossy compression?

Foundation knowledge for all subsequent image processing techniques

What is a Digital Image?

Comprehensive Definition

A digital image is a **2D array of pixels** (picture elements), where each pixel contains discrete intensity values representing color or brightness information at specific spatial locations.

Mathematical Foundation:

- Image function: $f(x, y)$
- Spatial coordinates: $x \in [0, M - 1], y \in [0, N - 1]$
- Intensity levels: $L = 2^k$ (k = bit depth)
- Pixel values: $0 \leq f(x, y) \leq L - 1$

Digital Characteristics:

- **Discrete**: Finite pixel locations
- **Quantized**: Limited intensity levels
- **Rectangular**: Grid-based sampling
- **Finite**: Bounded image dimensions

Common Bit Depths & Ranges:

- **1-bit**: Binary (0-1)
File size = $M \times N$ bits
- **8-bit**: Grayscale (0-255)
File size = $M \times N$ bytes
- **24-bit**: RGB Color (0-16.7M colors)
File size = $3 \times M \times N$ bytes

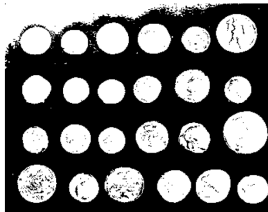
$$\text{File Size} = M \times N \times \text{bit depth} \div 8 \text{ bytes}$$

Four Main Image Categories

Grayscale Original
(scikit-image coins)



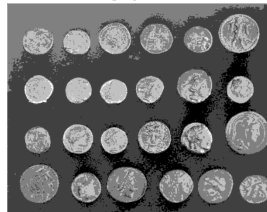
Binary Image
(Otsu threshold)



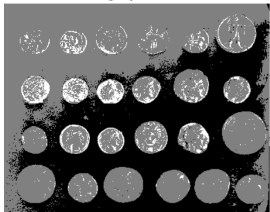
4-bit Quantized
(16 gray levels)



2-bit Quantized
(4 gray levels)



1-bit Quantized
(2 gray levels)



Color Image (RGB)
(scikit-image astronaut)



Red Channel
Only



Indexed Color
(16 colors, K-means)



1. Binary Images (1-bit)

- Values: 0 (black), 1 (white)

3. Color Images (24-32 bit)

- RGB: 8-bits per channel (16.7M colors)

Binary images contain only **two intensity values**: 0 (black) and 1 (white). Each pixel requires exactly **1 bit** of storage.

Technical Specifications:

- Pixel values: $f(x, y) \in \{0, 1\}$
- Storage: $\frac{M \times N}{8}$ bytes
- Memory efficiency: 8:1 vs grayscale
- Bit packing: 8 pixels per byte

Creation Methods:

- **Thresholding:** $f(x, y) = \begin{cases} 1 & \text{if } g(x, y) > T \\ 0 & \text{otherwise} \end{cases}$
- **Dithering:** Floyd-Steinberg algorithm
- **Edge detection:** Canny, Sobel operators

Applications:

- OCR systems
- Barcode reading
- Medical bone analysis
- Industrial inspection

Advantages:

- Fast processing
- Minimal storage
- Simple algorithms

Disadvantages:

- Loss of detail
- Limited applications
- Threshold sensitivity

Applications:

- Medical X-rays
- Satellite imagery
- Security systems
- Scientific imaging

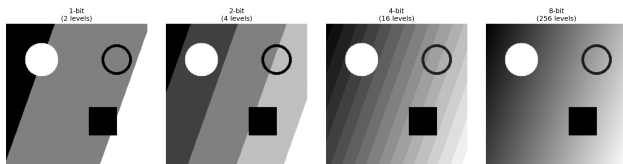
Storage: $M \times N$ bytes (8-bit)

Critical for scientific applications

Grayscale images represent intensity using shades of gray, typically **8-bit** (256 levels) or **16-bit** (65,536 levels).

Bit Depth Comparison:

- **8-bit:** 0-255, standard photography
- **12-bit:** 0-4095, digital cameras
- **16-bit:** 0-65535, scientific imaging
- **32-bit:** Floating point, HDR



RGB Conversion:

- **Luminance:** $Y = 0.299R + 0.587G + 0.114B$
- **Average:** $\text{Gray} = \frac{R+G+B}{3}$

RGB uses **additive color mixing** with Red, Green, Blue channels.
Standard: **24-bit color** (8 bits per channel).

Technical Specifications:

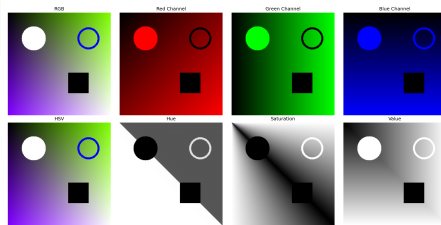
- Each channel: 0-255 (8-bit)
- Total colors: $256^3 = 16,777,216$
- Storage: $3 \times M \times N$ bytes
- Memory layout: Interleaved/planar

Extended Formats:

- **RGB565**: Mobile displays (16-bit)
- **RGB888**: Standard 24-bit
- **ARGB8888**: 32-bit with alpha
- **RGB48**: Professional (16-bit/channel)

Gamma Correction: $V_{out} = V_{in}^{\gamma}$ where $\gamma \approx 2.2$

Color Spaces & Channels



Advantage: Direct display compatibility
Challenge: Large file sizes

CMYK (Subtractive Model):

- Channels: Cyan, Magenta, Yellow, Black
- Application: Printing industry
- Color mixing: Ink absorption
- Smaller gamut than RGB

HSV/HSB (Perceptual Model):

- Hue: Color wheel (0-360°)
- Saturation: Purity (0-100%)
- Value: Brightness (0-100%)
- Intuitive color selection

Specialized Spaces:

- **YUV**: Video compression (luminance/chrominance)
- **LAB**: Perceptually uniform
- **XYZ**: CIE standard foundation

RGB to HSV:

$$V = \max(R, G, B)$$
$$S = \frac{V - \min(R, G, B)}{V}$$

Model Selection:

- **Display**: RGB color model
- **Print**: CMYK color model
- **Perception**: HSV color model
- **Compression**: YUV color model

Indexed color uses a **Color Lookup Table (CLUT)** with limited colors. Each pixel stores an **index** (8-bit) to the palette.

Technical Structure:

- Pixel values: 0-255 (indices)
- Palette: Up to 256 colors
- Storage: $M \times N + 256 \times 3$ bytes
- Effective when colors ≤ 256

Palette Algorithms:

- **Median Cut:** Recursive division
- **Octree:** Tree-based quantization
- **K-means:** Color space clustering

Applications:

- GIF animations
- Retro games
- Scientific visualization
- Embedded systems

Trade-offs:

- + Reduced file size
- + Fast palette swaps
- - Limited color range
- - Posterization effects

Modern usage: Specialized applications with acceptable color limitations

Image storage formats define how pixel data is organized, encoded, and stored in files. They determine **compression**, **metadata handling**, and **compatibility**.

Format Classification:

- **Raster vs Vector:** Pixel-based vs mathematical shapes
- **Compressed vs Uncompressed:** File size optimization
- **Lossless vs Lossy:** Quality preservation approach
- **Single vs Multi-frame:** Static images vs animations

Key Selection Criteria:

- **File Size:** Storage and bandwidth requirements
- **Quality:** Acceptable degradation level
- **Compatibility:** Software and device support
- **Features:** Transparency, animation, metadata
- **Processing Speed:** Encode/decode performance

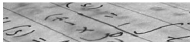
Common Format Categories:

- **Uncompressed:** BMP, PPM
Maximum quality, large files
- **Lossless Compressed:** PNG, GIF, TIFF
Quality preserved, moderate compression
- **Lossy Compressed:** JPEG, WebP
High compression, quality trade-offs
- **Professional:** RAW, TIFF, PSD
Maximum flexibility and quality

No single format is optimal for all applications
Context determines the best choice

Storage Formats - File Size Comparison

Storage Format Comparison - scikit-image Datasets

Camera Original 	PNG Lossless 	JPEG-90 PSNR: 40.3dB 	JPEG-50 PSNR: 32.6dB 	JPEG-10 PSNR: 28.4dB 
Coins Original 	PNG Lossless 	JPEG-90 PSNR: 42.3dB 	JPEG-50 PSNR: 31.0dB 	JPEG-10 PSNR: 26.3dB 
Checkerboard Original 	PNG Lossless 	JPEG-90 PSNR: 49.5dB 	JPEG-50 PSNR: 38.2dB 	JPEG-10 PSNR: 29.9dB 
Text Original 	PNG Lossless 	JPEG-90 PSNR: 40.8dB 	JPEG-50 PSNR: 35.1dB 	JPEG-10 PSNR: 29.6dB 

Windows Bitmap is Microsoft's native raster format, typically storing **uncompressed pixel data** with simple file structure.

Technical Specifications:

- File Structure: Header + Pixel data
- Color Support: 1, 4, 8, 16, 24, 32-bit
- Compression: Usually none (RLE available)
- Max Dimensions: Limited by 32-bit integers
- Byte Order: Little-endian (Intel format)

File Structure Details:

- **File Header:** 14 bytes (signature, file size)
- **Info Header:** 40+ bytes (dimensions, bit depth)
- **Color Palette:** Optional for ≤ 8 -bit images
- **Pixel Data:** Row-by-row, bottom-to-top
- **Row Padding:** Aligned to 4-byte boundaries

Advantages:

- Simple structure
- No compression artifacts
- Universal Windows support
- Fast loading

Disadvantages:

- Very large file sizes
- Limited web support
- No transparency support
- Minimal metadata

Use Cases:

- System icons
- Wallpapers
- Simple graphics
- Temporary image storage
- Legacy applications

Best for: Maximum quality when file size is not a concern

Joint Photographic Experts Group format uses **DCT-based lossy compression** optimized for photographic images with smooth color transitions.

Compression Pipeline:

1. **Color Space:** RGB $\rightarrow$ YCbCr conversion
2. **Subsampling:** Reduce chrominance resolution
3. **DCT Transform:** 8×8 block frequency analysis
4. **Quantization:** Discard high-frequency data
5. **Huffman Coding:** Entropy-based compression

Quality Control:

- Quality Setting: 1-100 scale (higher = better)
- Quantization Tables: Control frequency removal
- Compression Ratio: 10:1 to 50:1 typical
- Artifacts: Blocking, ringing, mosquito noise

Technical Features:

- **Progressive:** Interlaced loading capability
- **EXIF Data:** Camera metadata support
- **Color Profiles:** ICC profile embedding

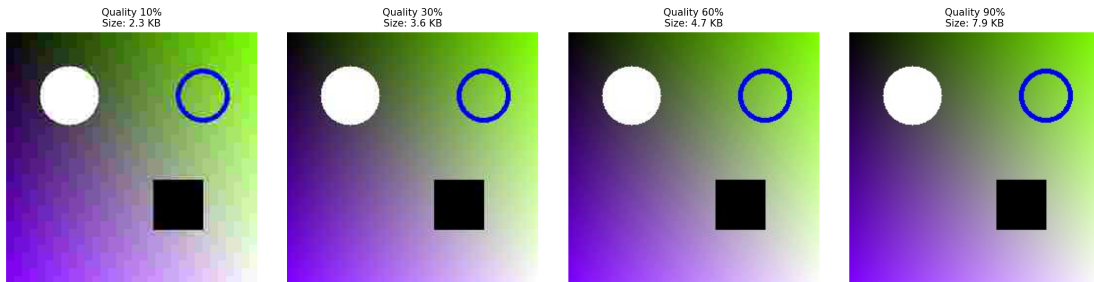
Advantages:

- Excellent compression ratios
- Universal support
- Adjustable quality
- Suitable for photographs

Disadvantages:

- Lossy compression
- Compression artifacts
- Poor for text/graphics
- No transparency

JPEG Compression - Quality Comparison



Quality vs File Size Trade-offs:

- **Quality 10%:** Smallest file, visible artifacts, suitable for thumbnails
- **Quality 30%:** Moderate compression, some artifacts, web use
- **Quality 60%:** Good balance, minimal artifacts, general photography
- **Quality 90%:** High quality, larger files, professional use

JPEG Compression Quality Comparison - scikit-image Dataset

Original Image
(scikit-image astronaut)



JPEG Quality 90
PSNR: 41.75 dB



JPEG Quality 50
PSNR: 34.65 dB



JPEG Quality 20
PSNR: 31.40 dB



JPEG Quality 5
PSNR: 25.90 dB



Compression Artifacts
(Difference from Original)



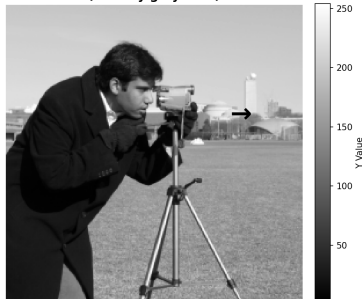
JPEG Compression - Step-by-Step Procedure

JPEG Compression Process using scikit-image "Camera" Dataset

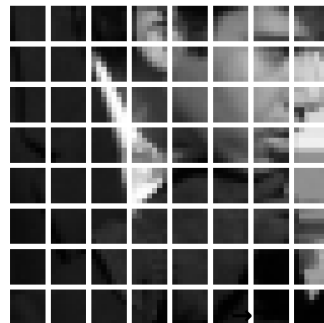
1. Original Image
(scikit-image camera)



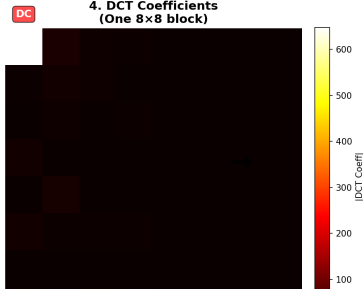
2. Luminance Component (Y)
(Already grayscale)



3. 8x8 Block Division
(64x64 section shown)



4. DCT Coefficients
(One 8x8 block)



5. After Quantization
(Information loss occurs)



6. Reconstructed Image
(After JPEG compression)



Portable Network Graphics provides **lossless compression** with advanced features like transparency and gamma correction, designed as a GIF replacement.

Technical Specifications:

- **Color Types:** Grayscale, RGB, Palette, Grayscale+Alpha, RGB+Alpha
- **Bit Depths:** 1, 2, 4, 8, 16 bits per channel
- **Compression:** DEFLATE algorithm (LZ77 + Huffman)
- **Interlacing:** Adam7 progressive display
- **Transparency:** Full alpha channel support

PNG Chunk Structure:

- **Critical Chunks:** IHDR, PLTE, IDAT, IEND
- **Ancillary Chunks:** tEXt, gAMA, cHRM, tIME
- **CRC Protection:** Error detection for each chunk
- **Extensible:** Custom chunk support

Compression Techniques:

- **Filtering:** Pre-processing for better compression
- **DEFLATE:** Dictionary-based compression
- **Palette Optimization:** Reduce colors when possible

Advantages:

- Lossless compression
- Full transparency support
- Robust error detection
- Wide format support

Disadvantages:

- Larger than JPEG for photos
- No animation support
- Limited metadata

PNG Compression Procedure (Lossless)

Step 1: Original Scanline

Pixel values:

120, 122, 121, 123, 125, 124, 126, 128

Size: 8 bytes

Step 2: Delta Filtering

Formula: $\text{pixel}[i] - \text{pixel}[i-1]$

Original:

120, 122, 121, 123, 125, 124, 126, 128

Filtered:

120, 2, -1, 2, 2, -1, 2, 2

Smaller values = better compression!

Step 3: DEFLATE Compression

1. LZ77 Algorithm:
 - Find repeated patterns
 - Replace with (distance, length)
2. Huffman Coding:
 - Frequent symbols → short codes
 - Rare symbols → long codes

Result: Compressed bitstream

Step 4: PNG File Structure

PNG Signature: 89 50 4E 47...

IHDR Chunk: Image header
- Width, height, bit depth
- Color type, compression

IDAT Chunk: Image data
- Compressed pixel data

Step 5: PNG Advantages

- ✓ Lossless compression
- ✓ Transparency support
- ✓ Good for graphics
- ✓ Cross-platform
- ✓ Patent-free

Best for:
• Screenshots
• Line art

Step 6: Typical Compression

Image Type	Compression
Screenshots	60-80%
Line art	70-90%
Photographs	20-40%
Graphics	50-70%

Note: Actual ratios depend on image content and complexity

Legacy format with **256-color limitation**, famous for animation support and LZW compression.

GIF Characteristics:

- Color Limit: 256 colors per frame
- Compression: LZW lossless algorithm
- Animation: Multiple frame support
- Transparency: 1-bit (on/off only)
- Interlacing: Progressive display

Applications:

- **GIF**: Web animations, simple graphics, logos
- **TIFF**: Professional photography, medical imaging, document archival, scientific data

Professional format supporting multiple compression methods, extensive metadata, and multi-page documents.

TIFF Features:

- Flexibility: Multiple compression options
- Color Support: 1-bit to 32-bit per channel
- Metadata: Extensive tag system
- Multi-page: Document scanning support
- Compression: None, LZW, ZIP, JPEG, PackBits

COMPRESSION

Fundamentals & Techniques

Optimizing Storage and Transmission

- **Compression Necessity:** Why compression is essential
- **Lossless Methods:** Perfect reconstruction techniques
- **Lossy Methods:** Quality vs. size optimization
- **Advanced Algorithms:** Modern compression standards

Image compression reduces file sizes by **eliminating redundancy** and exploiting **perceptual limitations** of human vision.

Types of Redundancy:

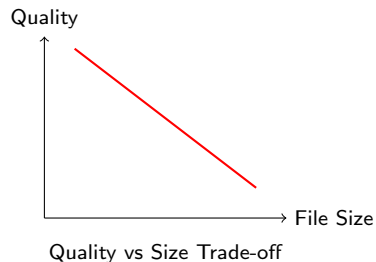
- **Spatial:** Neighboring pixels often similar
- **Spectral:** Color channels correlated
- **Temporal:** Video frame similarities
- **Psychovisual:** Human eye limitations

Compression Classification:

- **Lossless:** Perfect reconstruction possible
- **Lossy:** Some information permanently lost
- **Near-lossless:** Small controlled loss
- **Visually lossless:** Imperceptible quality loss

Performance Metrics:

- **Compression Ratio:** Original size / Compressed size
- **Bit Rate:** Bits per pixel (bpp)
- **PSNR:** Peak Signal-to-Noise Ratio
- **SSIM:** Structural Similarity Index



Lossless compression achieves **perfect reconstruction** by exploiting statistical redundancy without discarding information.

Huffman Coding:

- Principle: Variable-length codes based on frequency
- Algorithm: Build binary tree from symbol probabilities
- Efficiency: Optimal for symbol-by-symbol coding
- Applications: JPEG (AC coefficients), PNG
- Limitation: Requires probability knowledge

LZW (Lempel-Ziv-Welch):

- Principle: Dictionary-based pattern matching
- Algorithm: Build dictionary during encoding
- Advantage: Adaptive, no prior knowledge needed
- Applications: GIF, TIFF, PostScript
- Performance: Good for images with patterns

Run-Length Encoding (RLE):

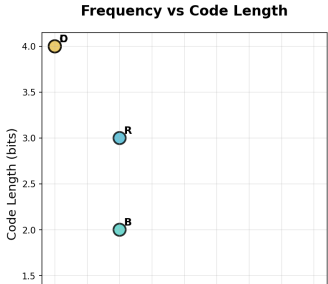
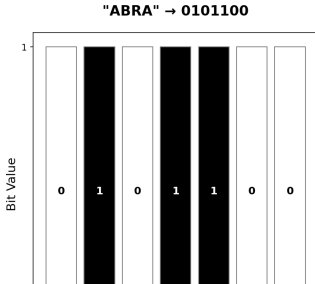
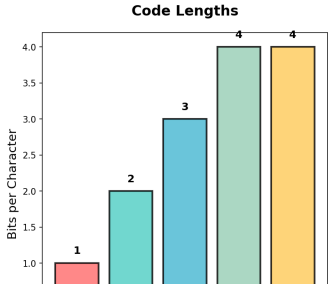
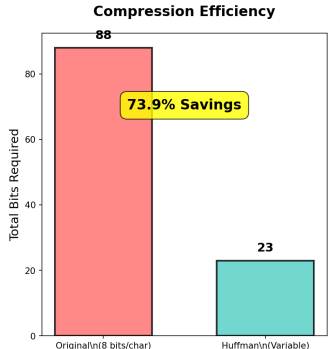
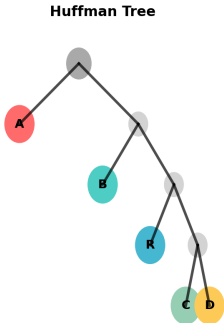
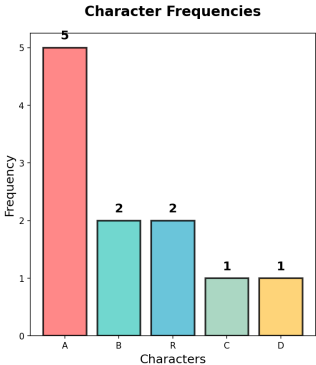
- Principle: Replace repeated values with count
- Example: AAABBBBCC → 3A4B2C
- Efficiency: Excellent for binary images
- Applications: BMP, PCX, some TIFF variants

Other Methods:

- **Arithmetic Coding:** Better than Huffman
- **DEFLATE:** LZ77 + Huffman (PNG)
- **LZMA:** Advanced dictionary compression

Perfect quality preservation
Moderate compression ratios

Huffman Encoding - Algorithm Demonstration



Huffman Encoding - Complete Procedure

Huffman Encoding Procedure

Step 1: Original Text

Text: ABRACADABRA

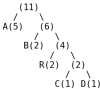
Length: 11 characters

ASCII bits: 88 bits

Step 2: Count Frequencies

A: 5
B: 2
R: 2
C: 1
D: 1

Step 3: Build Huffman Tree



Step 4: Assign Codes

A: 0
B: 10
R: 110
C: 1110
D: 1111

Step 5: Encode Text

A → 0
B → 10
R → 110
A → 0
C → 1110
A → 0
D → 1111
A → 0
B → 10
R → 110
A → 0
Encoded: 01011001110011110101100

Step 6: Results

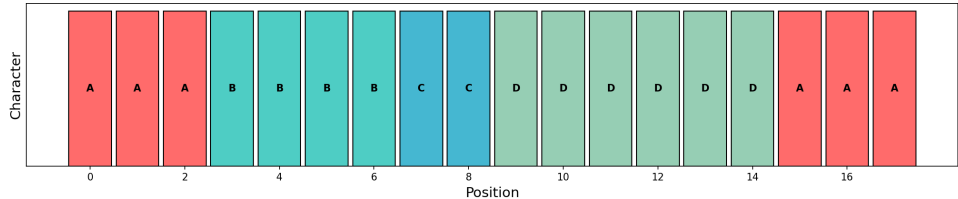
Original: 88 bits
Compressed: 23 bits
Savings: 65 bits

Compression Ratio:

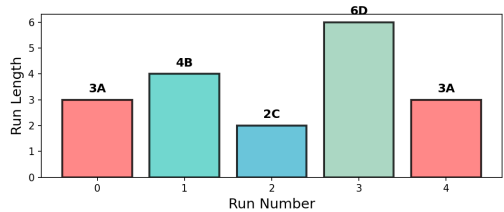
73.9%

Run Length Encoding - Simple but Effective

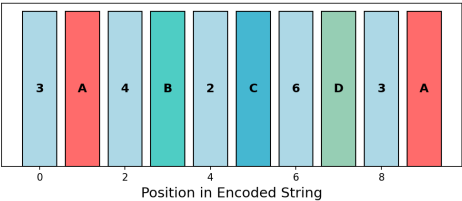
Original Data: "AAABBBBCCDDDDDDDDAAA"



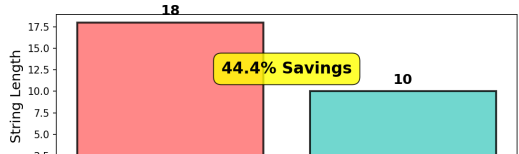
Identified Runs



RLE Encoded: "3A4B2C6D3A"



Compression Results



RLE Algorithm Steps

1. Scan left to right
2. Count consecutive chars
3. Write count + character
4. Repeat until end

Lossy compression achieves **high compression ratios** by permanently removing information that has **minimal perceptual impact**.

Transform-Based Compression:

- **DCT (Discrete Cosine Transform):** JPEG standard
- Principle: Convert spatial to frequency domain
- 8×8 Blocks: Process image in small segments
- Quantization: Remove high-frequency components
- Entropy Coding: Huffman encode remaining data

Wavelet-Based Compression:

- **JPEG 2000:** Modern wavelet standard
- Multi-resolution: Analyze multiple scales
- Better Quality: Less blocking artifacts
- Progressive: Quality/resolution scalability
- Applications: Medical, satellite imagery

Quantization Strategies:

- **Uniform:** Equal step sizes
- **Non-uniform:** Perceptually-based steps
- **Adaptive:** Content-dependent adjustment

Advantages:

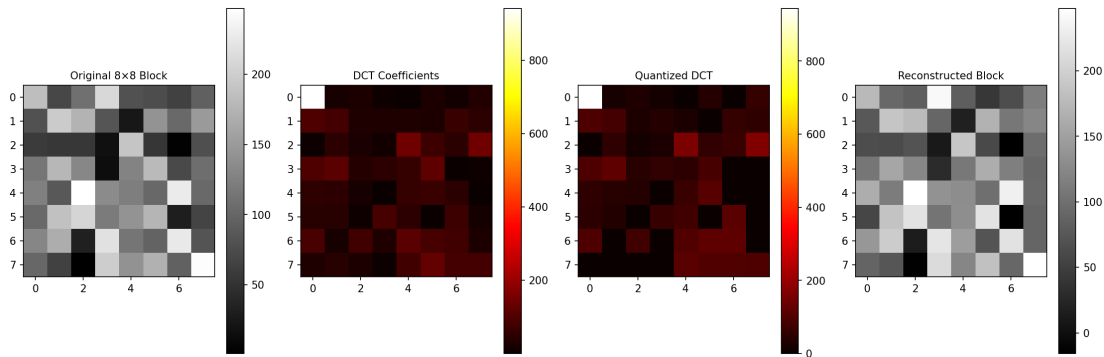
- High compression ratios
- Adjustable quality
- Suitable for photos

Disadvantages:

- Irreversible loss
- Compression artifacts
- Quality degradation

Trade-off: File size vs Visual quality

DCT (Discrete Cosine Transform) - JPEG Foundation



DCT Process Steps:

- **Spatial** → **Frequency**: Transform 8×8 pixel blocks to frequency domain
- **Energy Compaction**: Most energy concentrated in low frequencies (top-left)
- **Quantization**: Remove high-frequency details (human vision less sensitive)
- **Reconstruction**: Inverse DCT with some information loss

Contemporary image compression algorithms balance **compression efficiency**, **computational complexity**, and **visual quality** for specific applications.

Next-Generation Formats:

- **WebP**: Google's VP8-based format
- **HEIF**: High Efficiency (H.265-based)
- **AVIF**: AV1-based open standard
- **JPEG XL**: Latest JPEG committee standard

Performance Metrics Comparison:

- Compression Ratio: AVIF $\hat{>}$ HEIF $\hat{>}$ WebP $\hat{>}$ JPEG
- Quality (SSIM): JPEG XL $\hat{>}$ AVIF $\hat{>}$ HEIF $\hat{>}$ WebP
- Encode Speed: JPEG $\hat{>}$ WebP $\hat{>}$ HEIF $\hat{>}$ AVIF
- Decode Speed: JPEG $\hat{>}$ WebP $\hat{>}$ HEIF $\hat{>}$ AVIF
- Browser Support: JPEG $\hat{>}$ WebP $\hat{>}$ AVIF $\hat{>}$ HEIF

Specialized Applications:

- **Medical Imaging**: JPEG-LS, JPEG 2000 (lossless)
- **Satellite Data**: Custom wavelets, CCSDS standards
- **Mobile Photography**: HEIF with computational photography

Future Trends:

- AI-based compression
- Perceptual quality metrics
- Real-time optimization
- Edge computing integration

Evolution continues: Better quality at smaller sizes

Medical Imaging Systems:

- **DICOM Standard:** JPEG-LS, JPEG 2000 lossless
- Requirements: No diagnostic information loss
- Storage: Petabytes of patient data
- Transmission: PACS networks, telemedicine
- Example: 16-bit CT scans, 12-bit X-rays

Social Media Platforms:

- **Facebook:** WebP for web, HEIF mobile uploads
- **Instagram:** Aggressive JPEG compression with AI enhancement
- **TikTok:** H.264 video, JPEG thumbnails
- Challenges: Billions of images, mobile bandwidth
- Innovation: Neural super-resolution post-processing

E-commerce & Product Photography:

- **Amazon:** Multi-format delivery (WebP, JPEG, AVIF)
- Requirements: Fast loading, zoom capability
- Progressive Loading: Low-quality → high-quality
- Mobile Optimization: Responsive images, lazy loading

Scientific & Satellite Imagery:

- **NASA:** Custom wavelets, lossless compression
- **Google Earth:** Multi-resolution pyramids
- **Weather Services:** Real-time compression for radar

Compression enables the digital imaging revolution
From medical diagnosis to social media

Comprehensive understanding of digital image representation, storage formats, and compression techniques that form the foundation of modern digital imaging.

Core Concepts Mastered:

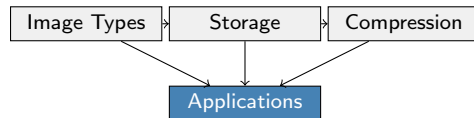
- **Image Types:** Binary, grayscale, color, indexed representations
- **Storage Formats:** BMP, JPEG, PNG, GIF, TIFF characteristics
- **Color Models:** RGB, YUV, color space conversions
- **Compression:** Lossless vs lossy trade-offs and algorithms

Critical Decision Framework:

- **Quality Requirements:** Medical = lossless, Web = acceptable loss
- **Storage Constraints:** Mobile vs server storage capabilities
- **Processing Power:** Real-time vs offline compression
- **Compatibility:** Universal support vs cutting-edge efficiency

Practical Guidelines:

- **Photography:** JPEG for sharing, RAW+lossless for archival
- **Graphics/Logos:** PNG for transparency, SVG for scalability
- **Web Development:** Progressive enhancement (AVIF→WebP→JPEG)
- **Scientific Data:** Always preserve original, compress copies



Questions & Discussion
From Theory to Real-World Applications